EFE KARAKOCA

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DESCRIPTION

Senior undergraduate student in Electrical and Electronics Engineering with strong interests in graph signal processing, network science, machine learning, and statistics.

EDUCATION

Boğaziçi University *2021–2027*
B.S. Electrical and Electronics Engineering GPA: 3.99/4.00
Senior

University of Illinois Urbana-Champaign *January 2025–May 2025*
Exchange Term: ECE Department GPA: 4.00/4.00
Junior

Boğaziçi University *2023–2027*
Double Major: Physics GPA: 4.00/4.00

Galatasaray High School *2016–2021*
Achieved the second-highest graduation ranking

PUBLICATIONS

E. Karakoca, S. Aviyente, "DyLNC: Dynamics-Aware Latent Network Component Decomposition for Temporal Networks,"*IEEE International Workshop on Machine Learning for Signal Processing (MLSP)*, 2026. **Accepted. [Paper]**

E. Karakoca, S. Aviyente, "MOSAIC: Modal Supra-Adjacency Learning for Multilayer Graphs with Inter-Layer Coupling,"*IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2027. **Submitted. [Paper]**

EXPERIENCE

Research with Prof. Selin Aviyente March 2026–Present
· Graph signal processing and graph topology learning
· Temporal and multi-view graph learning

Research with Prof. Ali Emre Pusane October 2025–Present
· Out-of-distribution robust neural receiver models for communication systems

Research with Prof. Zhi-Pei Liang March–August 2025
· Medical image processing with machine learning algorithms
· Clustering techniques, generative models, and dimension reduction
· Skull stripping models for MRI

Research with Dr. İlke Ercan June–November 2024
· Research on quantum circuits and transmon qubits
· Quantum algorithms

Internship at TÜBİTAK BİLGEM June–August 2024
· Radar technologies; worked on FMCW radars at BİLGEM

COURSE PROJECTS

Blind-Image Deblurring Fall 2025 - Image Processing

· Kernel estimation via posterior density learning and dictionary learning based optimization

Dirichlet Process in Clustering Perspective Fall 2025 - Bayesian Signal Processing

· Analysis of the Dirichlet Process and Chinese Restaurant Process from a clustering perspective

TECHNICAL SKILLS

Programming Languages	Python, MATLAB, C++
Machine /Deep Learning	Deep learning model architectures, PyTorch
Tools	LaTeX, Xilinx

RELEVANT COURSES

Core Courses
Signals and Systems
Probability Theory and Random Signals
Linear Algebra
Differential Equations
Complex Analysis
Numerical Analysis

Other Courses
Machine Learning
Digital Signal Processing
Bayesian Signal Processing
Digital Communications
Diffusion Magnetic Resonance Imaging
Statistical Physics
Image Processing

LANGUAGES

English	Advanced, TOEFL 110/120
French	Advanced, DALF C1
Turkish	Native

HONORS & AWARDS

İş Bankası Altın Gençler Prize 2021

Awarded for ranking 10th nationwide in the 2021 Turkish University Entrance Examination (YKS).